SoilGood — Test Cases (TS-01 to TS-10)

TC_DEVICE_001 — Happy Path

Related Scenario: TS-01 – Verify that adding / claiming a new IoT soil sensor on the farm works correctly.

Field	Details
Test Case ID	TC_DEVICE_001
Title	Claim sensor with a valid unused sticker code
Related Scenario	TS-01 – Verify that adding / claiming a new IoT soil sensor on the farm works correctly.
Preconditions	1. The SoilGood app is installed and can reach Supabase. 2. The farmer is signed in. 3. Profile, farm location, and soil texture onboarding are complete. 4. This farm has no linked device yet. 5. An unclaimed kit with claim_code SG-7K2M exists (farm_id null). 6. The user is on the Link your sensor screen.
Test Steps	1. Confirm the Claim code field is empty. 2. Tap the Claim code field. 3. Enter SG-7K2M. 4. Tap Link sensor. 5. Wait until the button loading indicator stops. 6. Check for an error banner. 7. Confirm the app opens the main shell (Home).
Test Data	Claim code: SG-7K2M
Expected Result	1. The claim succeeds. 2. No error banner is shown. 3. The app opens the main shell / Home. 4. The device farm_id is set to this farm. 5. The farmer never sees ingest_token.

TC_DEVICE_002 — Error Path

Related Scenario: TS-01 – Verify that adding / claiming a new IoT soil sensor on the farm works correctly.

Field	Details
Test Case ID	TC_DEVICE_002
Title	Reject claim when the sticker code was already used
Related Scenario	TS-01 – Verify that adding / claiming a new IoT soil sensor on the farm works correctly.
Preconditions	1. The farmer is signed in and on Link your sensor. 2. Profile and farm onboarding are complete. 3. This farm has no linked device yet. 4. Claim code SG-USED1 was already claimed by another farm.
Test Steps	1. Tap the Claim code field. 2. Enter SG-USED1. 3. Tap Link sensor. 4. Wait for the response. 5. Read the error banner. 6. Confirm the user is still on the claim screen.
Test Data	Invalid / edge data Claim code: SG-USED1
Expected Result	1. The claim fails. 2. Error banner shows: This claim code was already used. 3. No device is linked to this farm. 4. ingest_token is not shown. 5. The user remains on Link your sensor.

TC_AUTH_001 — Happy Path

Related Scenario: TS-02 – Verify that user registration, login, and logout work correctly.

Field	Details
Test Case ID	TC_AUTH_001
Title	Successful login with valid credentials then logout
Related Scenario	TS-02 – Verify that user registration, login, and logout work correctly.
Preconditions	1. The SoilGood app is accessible on a supported device. 2. A registered farmer account already exists. 3. The user is currently logged out on the Welcome Back screen. 4. Network is available.
Test Steps	1. Tap the Email Address field. 2. Enter a valid registered email. 3. Tap the Password field. 4. Enter the correct password. 5. Tap Sign in / Login. 6. Wait until navigation finishes (onboarding or app shell). 7. Open Profile. 8. Tap Logout / Sign out. 9. Confirm the login screen is shown again.
Test Data	Email: farmer@example.com Password: Farmpass1!
Expected Result	1. The user is authenticated successfully. 2. The app opens onboarding or the main shell. 3. Logout returns to the login screen. 4. The previous session can no longer open farm data without signing in again.

TC_AUTH_002 — Error Path

Related Scenario: TS-02 – Verify that user registration, login, and logout work correctly.

Field	Details
Test Case ID	TC_AUTH_002
Title	Login failure with invalid password
Related Scenario	TS-02 – Verify that user registration, login, and logout work correctly.
Preconditions	1. The SoilGood app is accessible. 2. A registered farmer account exists. 3. The user is logged out on the login screen. 4. Network is available.
Test Steps	1. Tap Email Address and enter the valid farmer email. 2. Tap Password and enter an incorrect password. 3. Tap Sign in / Login. 4. Wait for the response. 5. Read the error banner. 6. Confirm Home / shell did not open.
Test Data	Invalid / edge data Email: farmer@example.com Password: WrongPass999!
Expected Result	1. The login attempt fails. 2. A visible error message is shown. 3. The user remains on the login page. 4. No session grants access to farm soil or device data.

TC_ONB_001 — Happy Path

Related Scenario: TS-03 – Verify that farm onboarding (profile, location pin, soil texture / soil series) works correctly.

Field	Details
Test Case ID	TC_ONB_001
Title	Complete profile, farm pin, and soil texture onboarding
Related Scenario	TS-03 – Verify that farm onboarding (profile, location pin, soil texture / soil series) works correctly.
Preconditions	1. The farmer is newly signed in. 2. Profile, farm location, and soil texture are not yet saved. 3. Location permission can be granted when asked. 4. Network is available.
Test Steps	1. On Profile, enter first name and required fields. 2. Tap Continue / Save. 3. On Location, set the farm pin on the map (or use current location). 4. Save the farm location. 5. On Soil texture, select a texture and optional soil series. 6. Save and continue to device claim (or shell if claim is later).
Test Data	First name: Juan Soil texture: Clay Loam (or any listed value) Farm pin: valid lat/long in the Philippines
Expected Result	1. Each onboarding step saves successfully. 2. The farm row stores location and soil texture. 3. The user can proceed to the next required step. 4. No silent skip of required profile or pin.

TC_ONB_002 — Error Path

Related Scenario: TS-03 – Verify that farm onboarding (profile, location pin, soil texture / soil series) works correctly.

Field	Details
Test Case ID	TC_ONB_002
Title	Block continue when required onboarding fields are empty
Related Scenario	TS-03 – Verify that farm onboarding (profile, location pin, soil texture / soil series) works correctly.
Preconditions	1. The farmer is signed in on the Profile onboarding screen. 2. Required profile fields are empty. 3. Farm pin has not been set.
Test Steps	1. Leave First name empty. 2. Tap Continue / Save. 3. Note any validation message. 4. If Location can be opened, try to continue without a pin. 5. Confirm the app does not treat the farm as fully onboarded into Home.
Test Data	Invalid / edge data First name: (empty) Farm pin: (none)
Expected Result	1. Validation errors appear for required fields. 2. Onboarding does not complete. 3. The user cannot enter the main shell as a fully set-up farm. 4. No invented farm coordinates are saved.

TC_SOIL_001 — Happy Path

Related Scenario: TS-04 – Verify that ESP32 soil readings are correctly received, validated, and stored in Supabase.

Field	Details
Test Case ID	TC_SOIL_001
Title	Ingest a valid 8-in-1 soil reading into Supabase
Related Scenario	TS-04 – Verify that ESP32 soil readings are correctly received, validated, and stored in Supabase.
Preconditions	1. A claimed device exists with valid device_uid and ingest_token. 2. The ingest_soil_reading RPC is deployed. 3. Network between device/test client and Supabase is available.
Test Steps	1. Call ingest_soil_reading with in-range 8-in-1 values. 2. Wait for the RPC response. 3. Open soil_readings in Supabase (or refresh Home). 4. Confirm a new row exists for this device. 5. Confirm stored fields match the payload.
Test Data	moisture_percent: 45 ph: 6.5 salinity: 0.5 validation_status: ok Valid device_uid + ingest_token
Expected Result	1. The RPC succeeds. 2. One new soil_readings row is stored for the device. 3. Values match the sent payload. 4. The reading can become the latest on Home.

TC_SOIL_002 — Error Path

Related Scenario: TS-04 – Verify that ESP32 soil readings are correctly received, validated, and stored in Supabase.

Field	Details
Test Case ID	TC_SOIL_002
Title	Reject ingest with invalid device credentials
Related Scenario	TS-04 – Verify that ESP32 soil readings are correctly received, validated, and stored in Supabase.
Preconditions	1. The ingest_soil_reading RPC is available. 2. The tester can call the RPC with a wrong ingest_token. 3. Sensor values in the payload are otherwise in range.
Test Steps	1. Call ingest_soil_reading with a device_uid and wrong ingest_token. 2. Use otherwise valid moisture/pH values. 3. Observe the RPC error. 4. Confirm no new accepted reading was created from this forged call.
Test Data	Invalid / edge data device_uid: (any) ingest_token: wrong-token moisture_percent: 45
Expected Result	1. The RPC fails (invalid device credentials or equivalent). 2. No successful soil reading is stored for the forged call. 3. Failure is visible to the tester/logs — not a silent accept.

TC_HOME_001 — Happy Path

Related Scenario: TS-05 – Verify that Home shows the correct latest live soil condition and never silently shows fake or stale-as-fresh data.

Field	Details
Test Case ID	TC_HOME_001
Title	Home shows the newest soil reading as live truth
Related Scenario	TS-05 – Verify that Home shows the correct latest live soil condition and never silently shows fake or stale-as-fresh data.
Preconditions	1. The farmer is signed in with a claimed device. 2. At least two readings exist: older (moisture 20) and newer (moisture 42, pH 6.4). 3. The user opens the Home tab.
Test Steps	1. Open the Home tab. 2. Wait for soil cards / latest reading to load. 3. Note the moisture value shown. 4. Note the timestamp or reading identity if shown. 5. Confirm values match the newest reading, not the older one.
Test Data	Older reading: moisture 20 Newer reading: moisture 42, pH 6.4
Expected Result	1. Home displays the newer reading (moisture 42). 2. The older reading is not shown as current live truth. 3. Live 8-in-1 values match the newest row.

TC_HOME_002 — Error Path

Related Scenario: TS-05 – Verify that Home shows the correct latest live soil condition and never silently shows fake or stale-as-fresh data.

Field	Details
Test Case ID	TC_HOME_002
Title	Home empty or failed load never invents healthy soil values
Related Scenario	TS-05 – Verify that Home shows the correct latest live soil condition and never silently shows fake or stale-as-fresh data.
Preconditions	1. The farmer is signed in. 2. The farm has a claimed device but zero soil_readings rows, or soil fetch is forced to fail. 3. The user opens Home.
Test Steps	1. Open the Home tab. 2. Wait for first load to finish. 3. Check whether moisture/pH show invented numbers presented as live OK. 4. Check for empty or error messaging. 5. Pull to refresh and confirm the same rule holds.
Test Data	Invalid / edge data soil_readings: none Or simulated soil fetch failure
Expected Result	1. A visible empty or error state is shown. 2. No fake live moisture/pH is presented as normal healthy soil. 3. Pull-to-refresh does not invent default healthy values.

TC_WATER_001 — Happy Path

Related Scenario: TS-06 – Verify that irrigation / water today guidance is correct based on current soil + weather.

Field	Details
Test Case ID	TC_WATER_001
Title	Water today tip matches dry soil with low rain chance
Related Scenario	TS-06 – Verify that irrigation / water today guidance is correct based on current soil + weather.
Preconditions	1. The user is on Home with a latest soil reading. 2. Moisture is in a dry or critical_dry band. 3. Today rain probability is below the skip threshold (below 60%). 4. Farm pin exists so Open-Meteo weather can load. 5. Home Water today tip can load.
Test Steps	1. Open Home and wait for soil, weather, and Water today tip. 2. Note the moisture band / value. 3. Note rain chance on the forecast strip if shown. 4. Read the Water today / irrigation tip card. 5. Confirm the tip tells the farmer to irrigate (or clear dry guidance).
Test Data	Moisture band: dry or critical_dry Rain probability today: below 60%
Expected Result	1. Water today recommends irrigating (or urgent water-today guidance). 2. The tip is consistent with dry soil and low rain chance. 3. Guidance is tied to the current soil + weather context.

TC_WATER_002 — Error Path

Related Scenario: TS-06 – Verify that irrigation / water today guidance is correct based on current soil + weather.

Field	Details
Test Case ID	TC_WATER_002
Title	Do not urge watering when soil is critically wet
Related Scenario	TS-06 – Verify that irrigation / water today guidance is correct based on current soil + weather.
Preconditions	1. The latest reading moisture is in critical_wet or wet band. 2. The user is on Home with tips loaded.
Test Steps	1. Open Home with the wet reading as latest. 2. Wait for Water today / irrigation tip. 3. Read the tip carefully. 4. Confirm it does not tell the farmer to water today.
Test Data	Invalid / edge condition Moisture band: critical_wet (or wet)
Expected Result	1. Guidance says do not water / check drainage (or equivalent). 2. The app must not recommend irrigating while critically wet. 3. Wrong dry-day watering advice is not shown as current.

TC_ALERT_001 — Happy Path

Related Scenario: TS-07 – Verify that nutrient and soil out-of-band alerts fire correctly for critical readings.

Field	Details
Test Case ID	TC_ALERT_001
Title	Critical soil out-of-band reading creates a soil alert
Related Scenario	TS-07 – Verify that nutrient and soil out-of-band alerts fire correctly for critical readings.
Preconditions	1. The app shell is open so the notification evaluator can run. 2. The latest reading has a critical pH or salinity (or similar) band. 3. No soil_alert exists yet for this Manila day (first insert allowed).
Test Steps	1. Ensure a latest reading with critical out-of-band soil facts is present. 2. Allow the notification engine to evaluate. 3. Open Notifications (or check farm_notifications). 4. Confirm a soil_alert (or equivalent) appears. 5. Confirm severity is urgent when the band is critical.
Test Data	Example: ph_band critical_low Or salinity_band critical_high
Expected Result	1. A soil out-of-band alert is created or shown. 2. Title/body mention the soil issue. 3. Critical bands use urgent severity. 4. At most one of that type per farm per Manila day.

TC_ALERT_002 — Error Path

Related Scenario: TS-07 – Verify that nutrient and soil out-of-band alerts fire correctly for critical readings.

Field	Details
Test Case ID	TC_ALERT_002
Title	Normal in-band reading does not create a false soil alert
Related Scenario	TS-07 – Verify that nutrient and soil out-of-band alerts fire correctly for critical readings.
Preconditions	1. The latest reading has ok bands for pH, temp, EC, salinity, and nutrients. 2. The shell / evaluator can run. 3. The reading is fresh.
Test Steps	1. Load a fresh in-range reading with ok bands. 2. Run or wait for notification evaluation. 3. Check Notifications for soil_alert or nutrient_low. 4. Confirm none of those false alerts appear for this reading.
Test Data	Invalid / edge condition for false positive All soil/nutrient bands: ok Fresh recorded_at
Expected Result	1. No soil_alert is raised for a fully ok reading. 2. No nutrient_low alert is raised when N/P/K bands are ok. 3. The farmer is not falsely alarmed.

TC_OFFLINE_001 — Happy Path

Related Scenario: TS-08 – Verify that device-offline / no-new-reading detection works when the sensor stops sending.

Field	Details
Test Case ID	TC_OFFLINE_001
Title	Emit device_offline when latest reading is older than 2 hours
Related Scenario	TS-08 – Verify that device-offline / no-new-reading detection works when the sensor stops sending.
Preconditions	1. A claimed device exists. 2. Latest soil reading recorded_at is more than 2 hours ago. 3. The app shell is open. 4. No device_offline alert yet for this Manila day.
Test Steps	1. Confirm the latest reading timestamp is older than 2 hours. 2. Allow notification evaluation to run. 3. Open Notifications or inspect alert drafts. 4. Confirm device_offline appears.
Test Data	staleAfter: 2 hours recorded_at: now minus 3 hours
Expected Result	1. A device_offline alert is shown or saved. 2. Message indicates the sensor has not reported. 3. Dedupe is once per farm per Manila day.

TC_OFFLINE_002 — Error Path

Related Scenario: TS-08 – Verify that device-offline / no-new-reading detection works when the sensor stops sending.

Field	Details
Test Case ID	TC_OFFLINE_002
Title	Do not mark device offline when a fresh reading exists
Related Scenario	TS-08 – Verify that device-offline / no-new-reading detection works when the sensor stops sending.
Preconditions	1. Latest reading recorded_at is within the last 10 minutes. 2. The evaluator can run. 3. Soil bands may be ok.
Test Steps	1. Confirm the latest reading is fresh (about 10 minutes old). 2. Run notification evaluation. 3. Check for device_offline. 4. Confirm it is absent.
Test Data	Invalid / edge condition for false offline recorded_at: now minus 10 minutes
Expected Result	1. No device_offline alert is raised. 2. Fresh data is not treated as sensor-down. 3. The farmer is not told the device is offline incorrectly.

TC_RLS_001 — Happy Path

Related Scenario: TS-09 – Verify that only the authenticated farm owner can access their soil, farm, and device data.

Field	Details
Test Case ID	TC_RLS_001
Title	Signed-in owner can see own farm soil on Home
Related Scenario	TS-09 – Verify that only the authenticated farm owner can access their soil, farm, and device data.
Preconditions	1. User A owns Farm A with a claimed device. 2. At least one soil reading exists for that device. 3. The active session is User A.
Test Steps	1. Sign in as User A. 2. Open Home. 3. Confirm soil readings for Farm A appear. 4. Confirm the owner is not blocked from their own data.
Test Data	Session: User A Farm A soil reading present
Expected Result	1. User A can view Farm A soil on Home. 2. Device-linked readings for Farm A are visible. 3. Access matches the authenticated owner session.

TC_RLS_002 — Error Path

Related Scenario: TS-09 – Verify that only the authenticated farm owner can access their soil, farm, and device data.

Field	Details
Test Case ID	TC_RLS_002
Title	User B cannot see User A soil data
Related Scenario	TS-09 – Verify that only the authenticated farm owner can access their soil, farm, and device data.
Preconditions	1. User A has Farm A with soil readings. 2. User B has a separate account and farm. 3. The tester can sign in as User B.
Test Steps	1. Sign in as User B. 2. Open Home. 3. Confirm User A moisture/readings are not shown. 4. If any path accepts a farm id, attempt User A farm id and confirm deny or empty.
Test Data	Invalid / edge data Session: User B Target: Farm A readings
Expected Result	1. User B only sees own farm data (or empty). 2. User A readings are not visible. 3. No cross-farm data leak occurs.

TC_AI_001 — Happy Path

Related Scenario: TS-10 – Verify that AI recommendations are tied to the correct soil reading + weather snapshot and do not reuse outdated advice when inputs change.

Field	Details
Test Case ID	TC_AI_001
Title	Reuse saved Home AI when soil and weather story are unchanged
Related Scenario	TS-10 – Verify that AI recommendations are tied to the correct soil reading + weather snapshot and do not reuse outdated advice when inputs change.
Preconditions	1. Home already has a saved kind=home assessment. 2. It is tied to the current soil reading and weather fingerprint. 3. valid_until is still in the future. 4. The user is on Home.
Test Steps	1. Open Home and note Condition / Water today / Nutrients tip text. 2. Pull to refresh without changing soil or weather inputs. 3. Confirm the same saved tips remain (no unnecessary regen required). 4. Confirm tips still match the current reading / story.
Test Data	Same soil_reading_id Same weather story Assessment still valid
Expected Result	1. Saved Home AI is reused. 2. Tips stay tied to the current reading and weather. 3. No wrong tips from a different reading are shown as current.

TC_AI_002 — Error Path

Related Scenario: TS-10 – Verify that AI recommendations are tied to the correct soil reading + weather snapshot and do not reuse outdated advice when inputs change.

Field	Details
Test Case ID	TC_AI_002
Title	Do not keep old AI tips after a new soil reading changes the story
Related Scenario	TS-10 – Verify that AI recommendations are tied to the correct soil reading + weather snapshot and do not reuse outdated advice when inputs change.
Preconditions	1. Home shows tips for Reading Old (dry story). 2. A new Reading New arrives with a different band story (e.g. wet). 3. Home is open or refreshed so AI can recheck.
Test Steps	1. Note tips while Reading Old is latest. 2. Insert or receive Reading New with a different moisture story. 3. Wait for Home AI recheck or pull to refresh. 4. Confirm tips regenerate or update for Reading New. 5. Confirm dry-water advice is not left as current when soil is now wet.
Test Data	Invalid / edge condition Reading Old: dry Reading New: wet or critical_wet
Expected Result	1. Outdated dry-water advice is not left as current. 2. New tips match Reading New (or loading then new tips). 3. AI is not stuck on the old soil_reading_id story.